

ENTERPRISE PRODUCT REQUIREMENTS DOCUMENT

Brothers Mobile Shop ERP System

Complete Multi-Branch Enterprise Resource Planning, POS, Service Repair & Financial Management Platform

DOCUMENT VERSION

v1.0.0 (Production Blueprint)

TARGET STACK

React 18 + Vite 5 + Node.js 20 + MongoDB 7

ARCHITECTURE

Micro-Module Modular Monolith (28 Modules)

DATE & STATUS

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1. Executive Summary & Product Vision

Brothers Mobile Shop ERP is a purpose-built, high-performance Enterprise Resource Planning (ERP) platform designed specifically to meet the complex operational, financial, and inventory demands of mobile retail stores, authorized device repair centers, and mobile device wholesalers.

Core Value Proposition

Unlike generic retail ERPs, Mobile Shop ERP manages individual device serial numbers (IMEI tracking) through every stage of the product lifecycle: inventory intake, retail sale, wholesale distribution, warranty claim, and diagnostic repair. It combines point-of-sale efficiency with double-entry accounting and HR payroll automation.

1.1 Strategic Objectives

- **End-to-End Serial/IMEI Tracking:** Eliminate device fraud, inventory shrinkage, and warranty disputes with 15-digit IMEI lifecycle tracking.
- **Multi-Branch Governance:** Seamlessly handle multi-store branch inventory transfers, centralized pricing control, and aggregated financial reports.
- **Integrated Financial Accounting:** Automated posting from POS sales, purchase orders, repair bills, and expense vouchers to double-entry general ledgers.
- **Service & Repair Workflow:** Digital job sheets, diagnostic updates, technician assignments, and cost estimation for customer repairs.
- **Modern Dynamic Design:** Instant switching across 7 custom design themes (including Neumorphism, Glassmorphism, and Neo Brutalism) to match brand preferences.

2. User Personas & Role-Based Access Control (RBAC)

2.1 User Personas

1. Super Admin / Business Owner

Full system control, multi-branch revenue monitoring, audit log reviews, global settings, investor management, and financial statements.

2. Branch Manager

Manages store operations, branch inventory counts, staff attendance approvals, local sales oversight, and stock transfer requests.

3. POS Cashier / Sales Executive

Rapid checkout using barcode/IMEI scanner, invoice generation, customer due collection, thermal receipt printing, and sales return intake.

4. Stock & Inventory Specialist

Catalog maintenance, supplier purchase intake, IMEI serial registration, stock transfers dispatch/receive, and low-stock alerts.

5. Device Repair Technician

Repair job sheet diagnosis, labor & parts cost estimation, status updates (Received -> In Progress -> Repaired), and warranty checks.

6. Accountant & HR Manager

Journal entries, trial balance, P&L generation, loan/investor schedules, staff payroll processing, leave approvals, and pay slip printing.

2.2 Role-Based Permission Matrix (27 System Permissions)

Permission Scope	Super Admin	Branch Manager	POS Cashier	Stock Manager	Technician	Accountant
sales:create / sales:view	YES	YES	YES	NO	NO	VIEW
sales:refund	YES	YES	APPROVAL	NO	NO	NO
products:manage / imei:track	YES	YES	VIEW	YES	VIEW	VIEW
repairs:assign / repairs:edit	YES	YES	NO	NO	YES	NO
accounting:journal / reports:financial	YES	NO	NO	NO	NO	YES
payroll:process / leaves:approve	YES	LEAVE ONLY	NO	NO	NO	YES
users:manage / settings:edit	YES	NO	NO	NO	NO	NO

3. System Architecture & Tech Stack

3.1 Technological Stack Breakdown

Layer	Technology	Architectural Rationale
Client Framework	React 18 + Vite 5 (ESM)	Sub-second HMR development, optimal production bundle splitting, SPA speed.
State Management	Zustand + TanStack Query v5	Zustand for client theme & auth persistence; TanStack Query for server state caching.
UI Components	shadcn/ui + Tailwind v3 + AG Grid	Radix accessibility primitives, ERP-grade AG Grid data tables, dynamic theme engines.
Backend Server	Node.js 20 (ESM) + Express 4	Modular monolith pattern ('server/modules/'), native async/await.
Database Layer	MongoDB 7 + Mongoose 8	Document schema flex for IMEI attributes, fast indexing, atomic transaction sessions.
Real-Time Engine	Server-Sent Events (SSE) + EventEmitter	Lightweight unidirectional live updates for stock alerts & notifications without Socket.io overhead.
Validation & Auth	Zod + JWT (Bearer + HTTP-Only Cookie)	Strict runtime API validation, zero-trust token auth, TOTP MFA support.

3.2 Architectural Pattern: Modular Monolith

```
Mobile-Shop-ERP Architecture
├─ client/
│   ├── src/components/ui/      --> Tailwind + Radix Primitives
│   ├── src/pages/             --> Route Views (Dashboard, Sales, Accounting, etc.)
│   ├── src/store/             --> Zustand (authStore, themeStore)
│   └─ src/utils/              --> EventEmitter, offlineDB (IndexedDB sync)
├─ server/
│   ├── config/                --> DB, Env, Security, Swagger
│   ├── middleware/            --> Auth, Security Rate Limit, Zod Validator
│   ├── models/               --> Shared Models (AuditLog, RepairTicket)
│   └─ modules/               --> 28 Feature Modules (Routes, Controller, Service, Model,
Validator)
    ├── sale/                  --> POS Invoicing, Return, Wholesale
    ├── imei/                  --> Serial Tracker, Status Flow
    ├── repair/                --> Job Sheet, Diagnostic Billing
    └─ accounting/            --> Double-Entry Ledger, P&L, Balance Sheet
```

4. Comprehensive Module Breakdown (28 System Modules)

4.1 Sales & Point of Sale (POS) [`server/modules/sale`]

- **Fast Checkout POS:** Single-screen POS interface with instant barcode scanner integration and instant 15-digit IMEI lookup.
- **Multi-Payment Splits:** Supports Cash, Credit Card, Mobile Financial Services (MFS: bKash/Nagad), and Customer Credit/Due.
- **Invoice Generation:** Automatic generation of printable PDF invoices with custom store branding and thermal receipt printer support (80mm / 58mm).
- **Sales Return & Refund:** Restocking fee calculation, IMEI status restoration (`Sold` -> `In Stock`), and automated customer credit note issuance.

4.2 Wholesale Management [`server/modules/wholesale`]

- **Volume Pricing Tiers:** Custom volume discount matrices based on order quantity thresholds per product.
- **Credit Limit Validation:** Automated pre-sale checks against customer credit ceiling before wholesale order approval.
- **Bulk Serial Reservation:** One-click bulk selection and validation of IMEI lists for large wholesale dispatches.

4.3 Product Catalog & Inventory Master [`server/modules/catalog` & `product`]

- **Taxonomy Management:** Category, Brand, Model, and Variant hierarchies with custom attribute fields (RAM, Storage, Color).
- **Pricing & Profit Margins:** Purchase cost tracking, minimum retail selling price enforcement, and margin threshold warnings.
- **Barcode Management:** Internal barcode generation and sticker printing for accessories and non-IMEI inventory.

4.4 IMEI Serial Tracker & Lifecycle Passport [`server/modules/imei`]

- **Unique Serial Control:** Mandatory 15-digit IMEI registration for all mobile phone inventory intake.
- **Lifecycle Status Flow:** Strict state machine transitions: `In Stock` → `Sold` → `In Repair` → `Returned` → `Scrapped` .
- **IMEI History Passport:** Complete timeline audit showing vendor purchase invoice, stock movements, POS invoice, buyer profile, and service repair tickets.
- **Bulk Excel Intake:** High-speed CSV/Excel serial importer with instant checksum duplicate detection.

4.5 Multi-Branch Stock & Transfers [`server/modules/stock` & `branch`]

- **Branch Stock Isolation:** Real-time inventory visibility per store location.
- **Transfer Order Workflow:** Inter-branch stock transfer request, dispatch confirmation, in-transit status, and receiving verification.
- **Low-Stock Alerts:** Automated notification triggers when product quantity falls below defined safety stock thresholds.

4.6 Device Repair & Job Sheet Management [server/modules/repair]

- **Digital Job Sheets:** Ticket generation capturing device condition, passcode/pattern, physical flaws, and customer problem statement.
- **Status Progression:** Received → Diagnosing → In Progress → Repaired → Delivered → Cancelled .
- **Billing Integration:** Spare parts consumption from inventory and technician labor fees auto-posted to repair revenue ledgers.

4.7 Double-Entry Accounting & Financial Management [server/modules/accounting]

- **Chart of Accounts:** Hierarchical structure (Assets, Liabilities, Equity, Revenue, Expenses).
- **Automated Postings:** Sales invoices, wholesale orders, vendor purchases, repair bills, and payroll runs automatically generate journal entries.
- **Financial Reports:** Real-time General Ledger, Trial Balance, Profit & Loss Statement (P&L), and Balance Sheet generation.

4.8 HR, Attendance & Payroll [server/modules/employee , attendance , leave , payroll]

- **Attendance Tracking:** Daily check-in/check-out recording with overtime and late arrival computation.
- **Leave Approval Hierarchy:** Employee submission → Manager review → HR final approval with leave balance deduction.
- **Monthly Payroll Processing:** Itemized salary slip generation accounting for basic pay, bonuses, leave deductions, and advance adjustments.

4.9 Additional Enterprise Modules (Summary Matrix)

Module Name	Primary Functionality & Key Capabilities
Supplier Management	Vendor profiles, purchase orders, incoming serial intake, payable ledgers.
Customer CRM	Customer credit history, due collections, purchase frequency analytics.
Warranty Management	Warranty registration, duration tracking, claim replacement intake.
Expense Control	Petty cash vouchers, operational expense categories, receipt upload attachments.
Investor Management	Capital investment records, equity share percentages, dividend distribution.
Loan Management	Business loan tracking, interest schedule, repayment installment logs.
Document Vault	Encrypted storage for vendor contracts, employee IDs, and business licenses.
Real-Time SSE Engine	Server-Sent Events streaming live stock alerts, security logs, and repair tickets.
Audit Logging	Immutable security audit trail capturing user ID, IP address, action, and payload diff.
System Reports	Exportable analytics (PDF/Excel) for sales, inventory valuation, and top performers.

5. Data Models & Entity Schemas

5.1 Core Database Schemas (Mongoose / MongoDB 7)

1. Product Schema (`server/modules/product/product.model.js`)

```
const productSchema = new Schema({
  name: { type: String, required: true, trim: true, index: true },
  sku: { type: String, required: true, unique: true, uppercase: true },
  category: { type: Schema.Types.ObjectId, ref: 'Category', required: true },
  brand: { type: Schema.Types.ObjectId, ref: 'Brand', required: true },
  model: { type: Schema.Types.ObjectId, ref: 'Model', required: true },
  hasImei: { type: Boolean, default: true },
  costPrice: { type: Number, required: true, min: 0 },
  retailPrice: { type: Number, required: true, min: 0 },
  wholesalePrice: { type: Number, required: true, min: 0 },
  minStockLevel: { type: Number, default: 5 },
  isActive: { type: Boolean, default: true }
}, { timestamps: true });
```

2. IMEI Tracker Schema (`server/modules/imei/imei.model.js`)

```
const imeiSchema = new Schema({
  imeiNumber: { type: String, required: true, unique: true, length: 15, index: true },
  product: { type: Schema.Types.ObjectId, ref: 'Product', required: true },
  branch: { type: Schema.Types.ObjectId, ref: 'Branch', required: true },
  status: {
    type: String,
    enum: ['IN_STOCK', 'SOLD', 'IN_REPAIR', 'RETURNED', 'SCRAPPED'],
    default: 'IN_STOCK',
    index: true
  },
  purchaseOrder: { type: Schema.Types.ObjectId, ref: 'Purchase' },
  saleInvoice: { type: Schema.Types.ObjectId, ref: 'Sale' },
  warrantyExpiry: { type: Date }
}, { timestamps: true });
```

3. Sale Invoice Schema (`server/modules/sale/sale.model.js`)

```
const saleSchema = new Schema({
  invoiceNumber: { type: String, required: true, unique: true, index: true },
  branch: { type: Schema.Types.ObjectId, ref: 'Branch', required: true },
  customer: { type: Schema.Types.ObjectId, ref: 'Customer' },
  soldBy: { type: Schema.Types.ObjectId, ref: 'User', required: true },
  saleType: { type: String, enum: ['RETAIL', 'WHOLESALE'], default: 'RETAIL' },
  items: [{
    product: { type: Schema.Types.ObjectId, ref: 'Product' },
    imei: { type: Schema.Types.ObjectId, ref: 'IMEI' },
    quantity: { type: Number, default: 1 },
    unitPrice: { type: Number, required: true },
    subtotal: { type: Number, required: true }
  ]},
  subTotal: { type: Number, required: true },
  discount: { type: Number, default: 0 },
  taxAmount: { type: Number, default: 0 },
  grandTotal: { type: Number, required: true },
  paidAmount: { type: Number, required: true },
  dueAmount: { type: Number, default: 0 },
  paymentMethod: { type: String, enum: ['CASH', 'CARD', 'MFS', 'DUE', 'SPLIT'] },
}, { timestamps: true });
```

6. UI/UX Design Engine & 7 Custom Themes

The client application features a state-of-the-art Design System powered by Zustand (`client/src/store/themeStore.js`) allowing users to switch between 7 distinct UI modes in real time without reloading:

- 1. Flat Mode**

Clean, minimal corporate UI with crisp borders and subtle shadows for standard ERP usage.
- 2. Neumorphism**

Soft extruded tactile surfaces with dual inset/outset drop shadows creating 3D tactile depth.
- 3. Glassmorphism**

Translucent backdrop blur cards (``backdrop-filter: blur(12px)``) with frosted white borders.
- 4. Liquid Glass**

Fluid glass gradients with vibrant refraction effects and semi-transparent containers.
- 5. Neo Brutalism**

High-contrast, bold 3px black borders, vibrant solid colors, and retro hard offset box shadows.
- 6. Aurora**

Dynamic glowing background mesh gradients with modern dark-mode glowing accents.

7. Integrations & Real-Time Engine

- **Server-Sent Events (SSE):** Broadcasts instant notifications from ``server/modules/sse`` for low-stock alerts, invoice payouts, and repair completion.
- **Browser Event Emitter:** Decoupled client-side architecture using native browser event handling (``client/src/utils/EventEmitter.js``).
- **Client-Side PDF Generator:** High-performance receipt and invoice generation via ``jsPDF`` and ``jspdf-autotable``.
- **Offline Sync Architecture:** Service Worker + IndexedDB wrapper (``offlineDB.js`` & ``offlineSync.js``) queues offline transactions and syncs automatically when network reconnects.

8. Non-Functional Requirements (NFRs)

NFR Category	Specification Requirement	Verification Method
Performance	API response time < 100ms for 95% of queries. POS barcode lookup < 50ms.	Load testing with k6 / Autocannon.
Security	Bcrypt password hashing (salt=10), JWT token refresh flow, TOTP 2FA, Helmet headers, magic-byte file upload validation.	OWASP ZAP security scan & Biome code audit.
Data Integrity	Atomic MongoDB transaction sessions across multi-step POS sale & inventory stock updates.	Integration test suite.
Availability	99.9% uptime target. Offline POS sales caching via Service Worker.	Health monitoring & offline status checks.

9. DevOps & Deployment Architecture

The system is packaged for containerized deployment via Docker and Docker Compose:

```
# Production Deployment Architecture
services:
  mongodb:
    image: mongo:7
    ports: ["27017:27017"]
    volumes: [mongo_data:/data/db]

  server:
    build: ./server
    command: node server/server.js
    ports: ["5000:5000"]
    environment:
      - PORT=5000
      - MONGODB_URI=mongodb://mongodb:27017/mobile_shop_erp
      - JWT_SECRET=${JWT_SECRET}

  client:
    build: ./client
    ports: ["3000:80"]
```

10. Release Roadmap & Key Metrics

Phase	Target Features	Status / Timeline
Phase 1: Core ERP	Auth, Catalog, Product, IMEI Tracker, Stock, POS Sales, Accounting.	COMPLETED
Phase 2: Enterprise Ops	Multi-Branch, Repairs, HR/Payroll, Investor, Loan, SSE Stream, 7 Themes.	COMPLETED
Phase 3: Integrations	SMS Gateway OTP, Thermal Printer direct USB driver, Advanced BI dashboards.	Q4 2026 TARGET